INCH-POUND
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COMMERCIAL ITEM DESCRIPTION

Solids Aliphatic Polyurea Coating

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE

1. Scope. This commercial item description (CID) covers a 100% solid, two component aliphatic polyurea coating intended for use on cleaned, pretreated and primed (when required) substrates which include: metals like steel, galvanized steel, aluminum, plastic, composites, wood, fabric and concrete. The polyurea coating is intended to provide an extremely tough and durable coating that is impact resistant, stone and gravel chip resistant, abrasion resistant, chemical resistant, corrosion resistant and sound and vibration damping over a wide temperature range. ***This aliphatic polyurea coating is UV durable and contains no hazardous diphenyl methane diisocyante (MDI) species***.

This material is designed for use as a UV durable spray-on protective layer for thermal insulation and abrasion resistance on vehicle interior surfaces such as doors and floors and abrasion prone vehicle exterior surfaces such as wheel wells, under carriage, foot steps, equipment storage areas and beds for tactical, combat and construction equipment.

The material can be applied from a variety of application equipment ranging from high production volume plural component equipment, to moderate volume static mix cartridge guns and small volume spot repair kits. The appropriate film thickness is dependant on the particular application needs. A general guideline is to use approximately 50 mils of coating thickness in high use or chip prone areas like wheel well, undercarriage, and other similar areas and lower film thickness (approximately 20 mils) in less demanding application areas like vehicle interiors. The material can easily be repaired with small volume repair kits and readily bonds with itself.

2. SALIENT CHARACTERISTICS

2.1 Materials. The materials shall be used in accordance with the manufacturer's material specification for sprayable polyurea coating. The use of recovered material made in compliance with regulatory requirements is acceptable provided all requirements of this CID are met (see 4).

Beneficial comments, suggestions, or questions on this document should be addressed to U.S. Army Tank-automotive and Armaments Command, 6501 E. 11 Mile Road, Warren, MI 48397-5000 or emailed to standardization@tacom.army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at http://assist.daps.dla.mil.

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2.1.1 Materials deterioration, prevention and control. The material shall be a sprayable polyurea consisting of two components, both of which contain solids and resins that combine to form a 100% solid barrier coating.

2.1.2 Polyurea. The two liquid components cure through an exothermic reaction to form an elastomeric polyurea coating which shall meet the zero volatile organic compound requirement and be compatible with the chemical agent resistant coating (CARC) primers and topcoats per Table 1 of MIL- DTL-53072.

2.1.2.1.1 Resin component. The material shall be a polyamine based resin blend.

2.1.2.1.2 Characteristics of the resin component. The characteristics of the aliphatic polyamine based resin component shall be established by the following:

a. 100% solid, liquid resin
b. 8.1-8.3 lb. weight per gallon (gal.) and 0.984 kilograms (kg)per liter (L) specific gravity
c. Brookfield Viscosity 850-1350 cps (25°C)
d. polyamine based resin component

2.1.2.1 Isocyanate Component. The material shall be an aliphatic polyisocyanate ***that does not contain hazardous diphenyl methane diisocyante (MDI) species.***

2.1.2.1.3 Characteristics of the isocyanate component. The characteristics of the aliphatic isocyanate component shall be established by the following:

a. 100% solid, liquid resin
b. 9.0-9.2 lb. weight per gallon (gal.) and 1.091 kilograms (kg)per liter (L) specific gravity
c. Brookfield Viscosity 1800-2300 cps (25°C)
d. an aliphatic polyisocyanate (contains no hazardous MDI species)

2.1.2.3 Characteristics of the mixed components. The characteristics of the mixed components of 2.1.2.1.1 and 2.1.2.2.1 and tolerance ranges shall be established by the following:

a.	Hardness, Shore D†	60	ASTM D2240
b.	Tensile Strength (psi) †	3480	ASTM D412
c.	Elongation (%)	300-400	ASTM D412
d.	Flexural Modulus (psi) †	20,400 (±600)	ASTM D790
e.	Flexural Stress at 5% Strain (psi)	668 (±28)	ASTM D790
f.	Tabor Abrasion (mg) loss after 100 cycles CS10 wheel/500g load	0 mg	ASTM D4060
g.	Tear Resistance (lb.ft./in) †	549 (±26)	ASTM D624 Die C
h.	Toughness (psi) †	10,300	ASTM D412
i.	Youngs Modulus (psi)	32,633	ASTM D412

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j.	Tg	50°C	None

(Tested according to instrument manufacturer instructions.)

k.	Water Absorption, %	1.1% (24 hour)	ASTM D570
l.	Gel time	40 seconds	
m.	Tack free time	2 minutes	
n.	Minimum dry film thickness	20 mils	

o. Maximum dry film thickness unlimited
p. Mixed cured components classified as inert, no danger to the environment

†***Note: The higher hardness, tensile strength, flexural modulus and tear resistance values at similar % elongation, indicate this material provides increased toughness and ~~greate~~r energy adsorption.***

2.1.2.4 Optional additives. The polyurea coating shall allow for the addition of materials such as sand, garnet, quartz, slag, rubber, polymer particles or other aggregates for increasing non-skid performance uniformly throughout the coating or the top surface.

2.1.2.5 Color. The polyurea coating shall be able to be pigmented to any color requested as defined by FED-STD-595 (see 6.1.2). The color chip numbers used are as follows:

Color	Chip Number
Aircraft Black	37038
Aircraft Gray	36300
Aircraft Green	34031
Aircraft Insignia Blue	35044
Aircraft Red	31136
Aircraft White	37875
Aircraft Yellow	33538
Black	37030
Brown	383, 30051
Dark Green	34082
Dark Sandstone	33510
Earth Yellow	33245
Field Drab	33105
Green	383, 34094
Interior Aircraft Black	37031
Interior Aircraft Gray	36231
Olive Drab	34088
Sand	33303
Tan	686A, 33446

2.1.2.6 Flame resistance. The sprayed polyurea coating shall pass the Federal Motor Vehicle Safety Standards and Regulations (FMVSSR), Standard number 302, motor vehicle burn test with self-extinguishing (SE) rating (see 6.1.7).

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2.2 Configuration. The polyurea coating composition shall be applied by spraying and be compatible with CARC primers and topcoats per Army Drawing 12369000 (see 6.1.6).

2.3 Performance. Unless otherwise specified, the performance of the spray applied polyurea coating shall be verified by the following.

2.3.1 Adhesion. The pull off adhesion shall be 2500-5000 psi, per ASTM D4541, depending on substrate. Cross cut adhesion shall have a rating of 5 over steel and other suitable substrates as measured in accordance with ASTM D3359, Methods A and B (see 6.1.3).

2.3.2 Flexibility. The flexural modulus shall be 20,400 (±600) psi when tested according to ASTM D790. The % elongation shall be 300-400 % when tested according to ASTM D412 (see 6.1.3).

2.3.3 UV Durability. No chalking, fading, cracking and minimal (< 15% of initial) gloss loss after 4 years of Ft. Lauderdale, Florida exposure and 5000 KJ Xenon Weather-O-Meter.

2.3.4 GM9540P Accelerated Corrosion Testing. Testing shall be conducted for a period of more than 40 cycles, with 8 cycles approximating the equivalent of one year of “real world” exposure in this environment. After 40 cycles, the polyurea coating shall exhibit no scribe creep when applied to steel and aluminum (see 6.1.5).

2.3.5 Appearance. The cured coating shall have a uniform surface texture, free of sags, pits, blisters or visual defects.

2.3.6 Operational temperature range. The operational temperature range shall be from -40 to 150°F for continual exposure without material degradation or loss of physical properties and an operational range of -55 to 220°F for short, intermittent time intervals of 60 seconds without material degradation.

2.3.7 Dimensional Stability. (maximum shrinkage). The maximum shrinkage shall be < 1% in total.

2.3.8 Chip resistance. No hits to substrate and only slight surface scuffing using the SAE J400 Gravel-O-Meter test method using number 6 or 7 crushed stone, 5 cups of number 8 road gravel, and Hexagon nuts with panel temperatures at 25°C and -40°C (see 6.1.4).

2.3.9 Impact resistance. No damage with 500 gram weighted steel ball dropped from 100cm at 25°C and -40°C.

2.3.10 Chemical resistance. The sprayed and cured polyurea coating composition shall meet the following chemical resistance levels as in MIL-DTL-64159 (section 3.6.11).

a. Acids	Excellent
b. Bases	Excellent
c. Solvents	Excellent

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d. Oils	Excellent
e. Oxidizers	Excellent

2.3.11 Toxicity. The sprayed and cured polyurea coating shall contain no volatile organic compounds, or solvents of any type. The sprayed on polyurea coating shall have no adverse effects on the health of personnel when used as directed for the intended purpose.

2.4 Identification marking. Identification markings shall be in the form of a label affixed to the unit container and shall be legible and shall include the manufacturer's name, address, lot number, part number, contents verification and safety warning.

2.5 Unit container sizes. The two components of the polyurea coating shall be available in the following unit container sizes:

a. 300, 600 and 750 ml cartridges
b. 5 gallon (20 liter) containers
c. 55 gallon (208 liter) drums
d. 200, 300 and 400 gallon totes
e. 5000 gallon tank wagons

3. REGULATORY REQUIREMENTS.

The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4. PRODUCT CONFORMANCE.

The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards and quality assurance practice, and be the same product offered for sale in the commercial market place. The Government reserves the right to require proof of such conformance.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5. PACKAGING.

Preservation, packing and marking shall be specified in the contract or order.

6. NOTES.

AFRL Blast Resistance/Spall Liner Applications: This material has been tested on two separate occasions as a spall liner on a concrete block wall assembly by AFRL at Tyndall Air Force Base and in both instances the polyurea coating passed their requirements for blast and fragmentation resistance with a minimum of wall deflection. A small material and information kit was developed for the second test and the coating was successfully applied by AFRL personnel.

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The kit is portable and designed for remote field application using pre-filled disposable canisters. The kit is an "open and apply" design containing all necessary materials, supplies, and operating instructions which minimizes the need for application expertise and equipment upkeep.

6.1 Source of documents.

6.1.1 Code of Federal Regulations (CFR) is available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402 or website: www.access.gpo.gov.

6.1.2 FED-STD-595, "Colors Used in Government Procurement" is available from Defense Automation and Production Service, Building 4/D, 700 Robbins Avenue, Philadelphia, PA 19111-5094 or website: http://assist.daps.dla.mil/quicksearch

6.1.3 ASTM D2240, "Standard Test Methods for Rubber Property Durometer - Hardness"; ASTM D412, "Standards Test Methods for Vulcanized Rubber and Thermoplastic Elastomers-Tension"; ASTM D790, "Standard Test Method for Flexural Properties of Unreinforced and Reinforced and Electrical Insulating Materials"; ASTM D4060, "Standard Test Method for Abrasion of Organic Coatings by the Taber Abrader"; ASTM D624, "Standard Test Method for Tear Strength of Conventional Vulcanized Rubber and Thermoplastic Elastomers"; ASTM D570, "Standard Test Method for Water Adsorption of Plastics"; ASTM D3359, "Standard Test Method for Measuring Adhesion by Tape Test"; ASTM D4541, "Standard Test Method for Pull-Off Strength of Coatings Using Portable Adhesion Testers" are available from ASTM International, P.O. Box C700, 100 Barr Harbor Drive, Conshohocken, PA 15096-2959 or website: www.astm.org.

6.1.4 SAE J400, "Test for Chip Resistance of Surface Coat", is available from the Society of Automotive Engineers Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001 or website: www.sae.org.

6.1.5 GM 9540P, "Acceleration Corrosion Test"; is available from General Motors Corporation c/o Global Engineering Documents, 15 Inverness Way East, Inglewood, CO 80112.

6.1.6 Army Drawing 12369000, "Painting System Index" is available from the Contracting Officer, U.S. Army Tank-Automotive and Armaments Command, 6501 E. 11 Mile Road, Warren, MI 48397-5000.

6.1.7 Standard number 302, "Flammability of Interior Materials" is available from the Federal Motor Vehicle Safety Standards and Regulations (FMVSSR), U.S. Department of Transportation, 400 7th Street S.W., Room 6111, mail code: NSA-30, Washington. D.C. 20590 or website: http://www.nhtsa.dot.gov/cars/rules/import/FMVSS/

6.2 Toxicity. If a Material Safety Data Sheet (MSDS) is required by law, it is to be shipped with the container. Contracting officers will identify those activities requiring copies of completed MSDS's prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313; and 29 CFR 1910.1200 requires that the MSDS for each hazardous chemical used in an operation must be readily

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available to personnel using the material. Contracting officers will identify the activities requiring copies of the MSDS.

6.3 Ordering data. The contract or order shall specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Packing requirements

6.4 Key words.

Abrasion protection
Corrosion protection
Corrosion resistance
Impact protection
Chip resistant
Sound damping
Vibration damping
UV durable
Tough

Custodian:
Army - AT

Preparing Activity:
Army – AT

Project Number 8030-2007-008

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